# Complex Numbers: Roots of Unity

Questions 1–30

**Question 1.** Convert the following points into the form $a + b\mathrm{i}$ where $a$ and $b$ are real constants:

a) $\mathrm{e}^{\frac{\pi}{2}\mathrm{i}}$
b) $\mathrm{e}^{\frac{3\pi}{4}\mathrm{i}}$
c) $\mathrm{e}^{\pi\mathrm{i}}$
d) $\mathrm{e}^{\frac{3\pi}{2}\mathrm{i}}$
e) $\mathrm{e}^{\frac{7\pi}{6}\mathrm{i}}$
f) $\mathrm{e}^{2\pi\mathrm{i}}$

**Question 2.**

a) Find the three roots of the equation $z^3 = 1$ giving your answers in cartesian form.
b) Plot the three roots on an Argand diagram.
c) State the name of the polygon formed by the points representing the roots.
d) Find the sum of the roots.
e) Find the area of the shape formed by the roots giving your answer in exact form.

**Question 3.**

a) Find the five roots of the equation $z^5 = 1$ giving your answers in exponential form.
b) Plot the five roots on an Argand diagram.
c) State the name of the polygon formed by the points representing the roots.
d) Find the sum of the roots.
e) Find the area of the shape formed by the roots giving your answer to three significant figures.

**Question 4.**

a) Find the six roots of the equation $z^6 = 1$ giving your answers in cartesian form.
b) Plot the six roots on an Argand diagram.
c) Find the sum of the roots.
d) Find the area of the shape formed by the roots giving your answer in exact form.

**Question 5.**

a) Find all the roots of the equation $z^7 = 1$ giving your answers in exponential form.

b) Plot the seven roots on an Argand diagram.

c) Find the sum of the roots.

d) Find the area of the shape formed by the roots giving your answer to three significant figures.

**Question 6.** Given that $z^n = 1$ for positive integer $n$

a) Find the area of the shape formed by the roots giving your answer in terms of $n$

b) Find the area of the shaped formed by the roots as $n \to \infty$

**Question 7.** Prove that the distance between any two consecutive $n^{\text{th}}$ roots of unity is $2\sin\left(\frac{\pi}{n}\right)$

**Question 8.** An equilateral triangle is centred at the origin with one vertex at $(2, 0)$

a) Find the coordinates of the other vertices.

b) Find the area of the triangle giving your answer in exact form.

**Question 9.** A regular hexagon is centred at the origin with one vertex at $(3, 0)$

a) Find the coordinates of the other vertices.

b) Find the area of the hexagon.

**Question 10.** Show that if $z$ is an $n^{\text{th}}$ root of unity then its complex conjugate is also an $n^{\text{th}}$ root of unity.

**Question 11.** Prove that the sum of the $n$ roots of unity is zero.

**Question 12.** Show that

a) $\omega = \cos\frac{4\pi}{5} + \mathrm{i}\sin\frac{4\pi}{5}$ is a solution to $z^5 = 1$

b) $\omega = \mathrm{e}^{\frac{2\pi}{7}\mathrm{i}}$ is a root of the equation $z^7 = 1$

**Question 13.**

a) Find the five roots of the equation $z^5 - 1 = 0$ giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \pi$

b) Hence show that

$$\cos\left(\frac{2\pi}{5}\right) + \cos\left(\frac{4\pi}{5}\right) = -\frac{1}{2}$$

**Question 14.**

a) Find the seven roots of the equation $z^7 - 1 = 0$ giving your answer in the form $r(\cos\theta + \mathrm{i}\sin\theta)$ where $-\pi < \theta \leq \theta$

b) Hence show that

$$\cos\left(\frac{2\pi}{7}\right) + \cos\left(\frac{4\pi}{7}\right) + \cos\left(\frac{6\pi}{7}\right) = -\frac{1}{2}$$

**Question 15.** By considering the fifth roots of unity, prove

$$\cos\left(\frac{\pi}{5}\right) + \cos\left(\frac{3\pi}{5}\right) = \frac{1}{2}$$

**Question 16.** Derive a cubic polynomial with integer coefficients, expressed in expanded form, whose roots are:

a) the three cube roots of unity.

b) the midpoints on the Argand diagram of the three cube roots of unity.

**Question 17.** It is given that

$$f(z) = z^5 + z^4 + z^3 + z^2 + z + 1$$

a) Find all of the solutions to $f(z) = 0$
b) Hence fully factorise $f(z)$

**Question 18.** $f(z) = z^7 + z^6 + z^5 + z^4 + z^3 + z^2 + z + 1$

a) Find all of the solutions to $f(z) = 0$
b) Hence fully factorise $f(z)$

## Working with $\omega$

**Question 19.** If $\omega$ is a complex cube root of unity and $\omega \neq 1$, without directly using the roots of unity simplify:

a) $\omega^3$

b) $\omega^5$

c) $\omega^8 - \omega^5 + \omega^6$

d) $\frac{1}{\omega}$

**Question 20.** Given $\omega$ is an $n^{\text{th}}$ root of unity, by considering the roots in their exponential form,

a) show that the complex conjugate of $\omega$ is $\omega^{n-1}$

b) Hence, find the complex conjugate of $\omega^k$ for integer $k < n$

**Question 21.** Let $\omega = \mathrm{e}^{\frac{\pi}{3}\mathrm{i}}$ where $\omega$ is a root of unity, find

a) $\mathrm{Re}(\omega + \omega^2 + \omega^3 + \omega^4 + \omega^5)$

b) $\mathrm{Im}(\omega^2 + \omega^4)$

**Question 22.** Let $\omega = \mathrm{e}^{\frac{\pi}{4}\mathrm{i}}$ where $\omega$ is a root of unity, find

a) $\mathrm{Re}(\omega^2 + \omega)$

b) $\mathrm{Im}(1 + \omega^4)$

**Question 23.** Given that $\omega$ is a complex $4^{\text{th}}$ root of unity such that $\omega \neq 1$, show that

$$(1-\omega)(1-\omega^2)(1-\omega^3) = 4$$

**Question 24.**

a) Simplify $1 + x + x^2 + \cdots + x^{n-1}$

b) Given $\omega$ is a principal $n^{\text{th}}$ root of unity so that $z^n - 1 = 0$, show that

$$\prod_{k=1}^{n-1}(1-\omega^k) = n$$

**Question 25.**

a) Prove that $|z_1 + z_2|^2 = (z_1 + z_2)(z_1^* + z_2^*)$

b) Given that $\omega$ is a cube root of unity, show that

$$|\omega - \omega^2|^2 = 2 - (\omega + \omega^*)$$

**Question 26.** If $\omega$ is a complex cube root of unity and $\omega \neq 1$, without directly using the roots of unity find

a) $(1 - \omega)(1 - \omega^2)$

b) $(1 - \omega)(1 - \omega^2)(1 - \omega^3)$

c) $(1 + \omega)(1 + \omega^2)$

d) $(1 + \omega)(1 + \omega^2)(1 + \omega^3)$

**Question 27.** If $\omega$ is a complex cube root of unity such that $\omega \neq 1$, and $a$ and $b$ are real constants, find each of the following in terms of $a$ and $b$:

a) $(a - b\omega)(a - b\omega^2)$
b) $(a - b\omega)(a - b\omega^2)(a - b\omega^3)$
c) $(a + b\omega)(a + b\omega^2)$
d) $(a + b\omega)(a + b\omega^2)(a + b\omega^3)$

**Question 28.** If $\omega$ is a complex $4^{\text{th}}$ root of unity and $\omega \neq 1$, without directly using the roots of unity find

a) $\omega(1 + \omega + \omega^2)$
b) $\omega(1 + \omega)(1 + \omega^2)$
c) $\left(\omega + \frac{1}{\omega}\right)^2 + 2\left(\omega + \frac{1}{\omega}\right)$

**Question 29.** If $\omega$ is a complex $3^{\text{rd}}$ root of unity and $\omega \neq 1$,

a) find $\omega + \frac{1}{\omega} + 1$

b) Hence, using your result from part a), prove that $\cos \frac{4\pi}{3} = -\frac{1}{2}$

**Question 30.** If $\omega$ is a complex $5^{\text{th}}$ root of unity and $\omega \neq 1$,

a) find $\left(\omega + \frac{1}{\omega}\right)^2 + \left(\omega + \frac{1}{\omega}\right)$

b) Hence show that

$$\cos \frac{4\pi}{5} = \frac{p + q\sqrt{5}}{r}$$

where $p$, $q$ and $r$ are constants to be found.